(Wednesday) 15/10/2025 (Class right after recess, no Monday lecture)

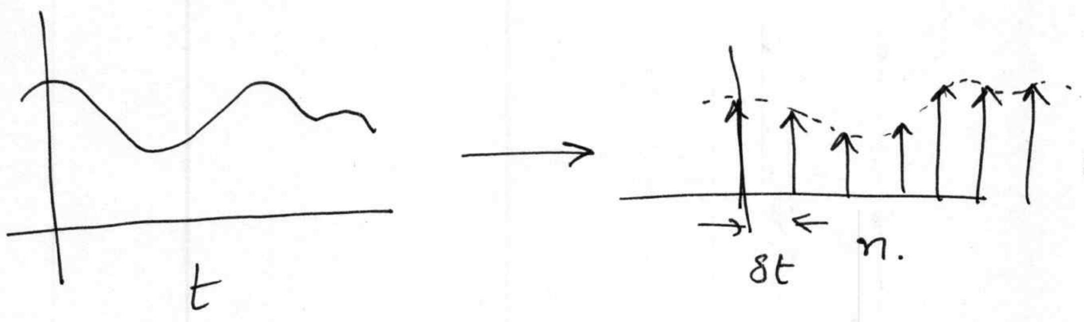
ADC: Sampling, Quantization and Reconstruction.

Topic included in the Midterm

2-7 Fundamentals of analog-to-digital-Conversion and Vice-Versa.

- Two parts to this conversion.

  (1) Sampling: Continuous to ~~real~~ discrete time on the x-axis of the signal.



The sampled signal takes on continuous values at rational numbers.

$$f: \mathbb{R} \to \mathbb{R} \longrightarrow \boxed{\text{Sampling}} \longrightarrow g: \mathbb{Q} \to \mathbb{R}.$$

$$t \mapsto f(t) \qquad\qquad \cancel{q \mapsto g(nq)}$$

or $$g[n] = f(n\delta t)$$

(2) Quantization: Discretization in magnitude on the y-axis.

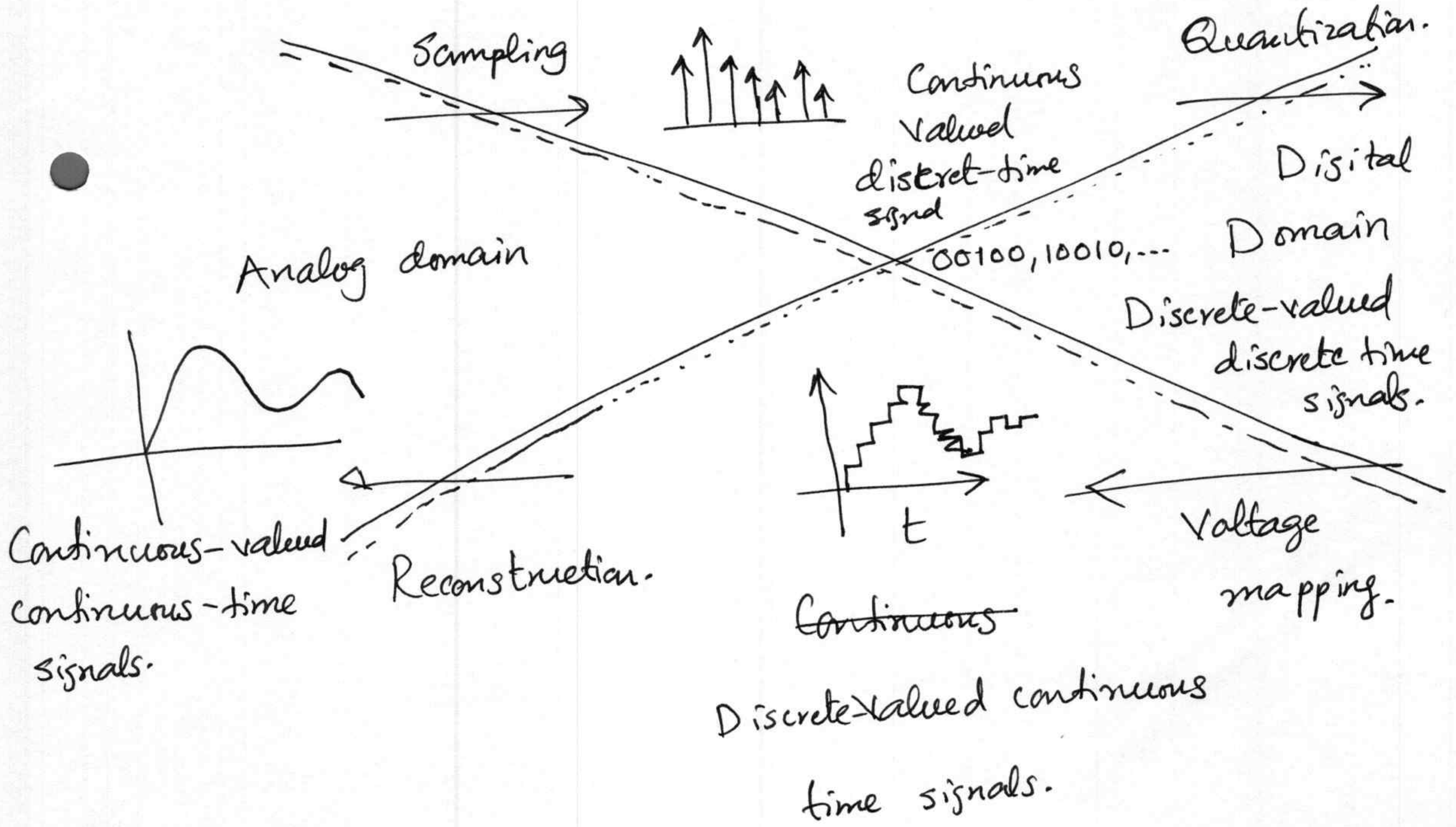
The continuous values are mapped to rational ones.

$$g: \mathbb{Q} \to \mathbb{R} \to \boxed{\text{Quantizer}} \to h: \mathbb{Q} \to \mathbb{Q}.$$

$$\cancel{q \mapsto h[nq] \in \mathbb{Q}}.$$

$$g[n] = [\![g[n]]\!]_{\mathbb{Q}}.$$

Some sources produce analog signals, such as speech and audio.



ADC: DV-DT (Discrete-valued, discrete-time signal)

$\uparrow \downarrow$*

CV-DT. (Continuous Valued, Continuous time signal).

DAC: Inverse of ADC

ADC has typically 3 subsystems:

1. (AAF): Anti-aliasing Filter.
2. Sampler: $x[nT_s] = x(t)\big|_{t=nT_s}$ T_s: sampling period.
3. Quantizer: $x_q[n] = Q\big[x[nT_s]\big]$

Figure 2-15: High-level block diagram of an ADC.

2.7.1 — Sampling

Sampling a signal $x(t)$ at a sampling period T_s.

$$x[nT_s] = x(t)\big|_{t=nT_s}.$$

Sampling can also be represented as an inner product with a Dirac delta, at the sampling nT_s.

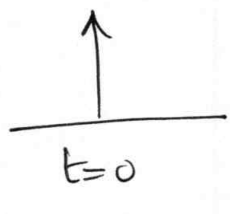
$$x[nT_s] = \langle x(t), \delta(t-nT_s) \rangle$$

$$= \int_{-\infty}^{+\infty} x(t)\, \delta(t-nT_s)\, dt.$$

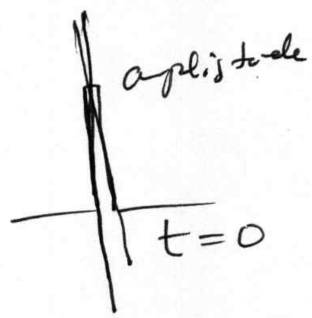
Main parameter of sampler: $f_s = \frac{1}{T_s}$ sampling freq.

Most samplers operate on voltage, some can operate on current.

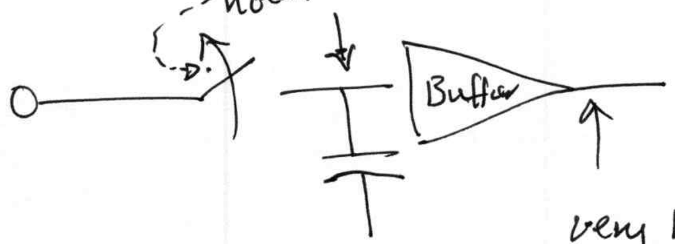
Conceptually:



Practically:



Sampling is typically done by sample-and-hold (S/H) circuits. The purpose of the circuit is to keep the input signal constant during ADC conversion.



The capacitor remains in hold mode for the duration of the sampling period T_s.

The impulse response of the ideal S/H circuit is:

$$h(t) = u(t) - u(t - T_s)$$
$$= \Pi\left(\frac{t}{T_s} - \frac{1}{2}\right)$$

and the frequency response is

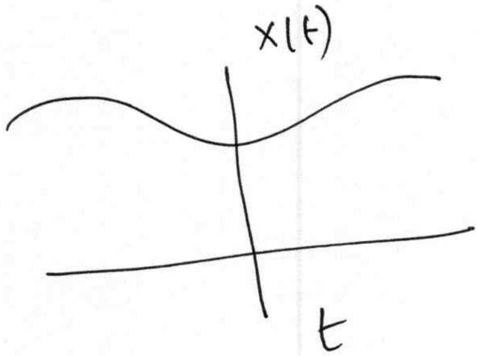
$$H(f) = T_s \frac{\sin \pi f T_s}{\pi f T_s} e^{-j\pi f T_s}.$$

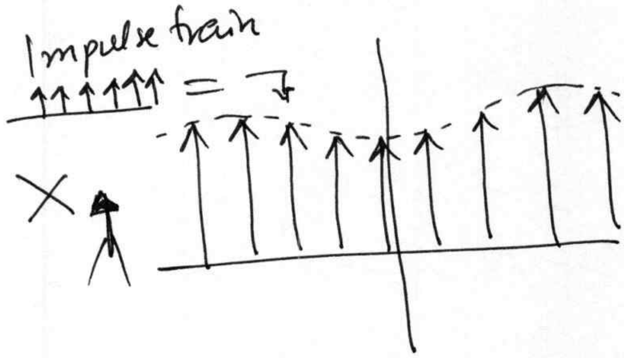
Suppose i/p signal: $x(t) \longleftrightarrow X(f)$, and the o/p of the sampler is a sampled signal $x_s(t)$. The sampled signal can be viewed as a product:

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$\sum_n \delta(t-nT_s) \longrightarrow$ impulse or a sampling function.

This give $$x_s(t) = \sum_n x(t)\,\delta(t-nT_s)$$
$$= \sum_n x[nT_s]\,\delta(t-nT_s)$$





Example 2-11 Fourier transform of a pulse train.

Comput $\mathcal{F}\left\{\sum_n \delta(t-nT)\right\}$

$$\sum_n \delta(t-nT_s) \overset{\mathcal{F}}{\longleftrightarrow} \sum_n e^{-j2\pi f n T_s}$$

Q for exam $\longrightarrow$ But FT of an impulse train is not unique!

There is another expression too for

$$\mathcal{F}\left\{\sum_n \delta(t-nT_s)\right\} \longrightarrow$$

Impulse train $\longleftrightarrow \begin{cases} \sum_n e^{j2\pi f nT} \\ \boxed{\frac{1}{T}\sum \delta\left(t - \frac{n}{T}\right)} \end{cases}$

This expression is more useful for the sampling theorem.

The Fourier Transform of a product of two functions is equal to the convolution of ~~of~~ their respective Fourier transform.

Thus,

$$X_s(f) = X(f) * \frac{1}{T_s}\sum_n \delta\left(f - \frac{n}{T_s}\right)$$

$$= \frac{1}{T_s}\sum_n \underline{X}\left(f - \frac{n}{T_s}\right)$$

$$= \frac{1}{T_s}\left\{ \cdots + \underline{X}(f+f_s) + \underline{X}(f) + \underline{X}(f-f_s) + \cdots \right\}$$

$$= f_s\underline{X}(f) + \sum_{\substack{n=-\infty \\ n\neq 0}}^{n=\infty} \underline{X}(f - nf_s)$$

"We see that the spectrum of the sampled signal is periodic with period in frequency equal to the sampling frequency.

$f_s = 1/T_s$"

Infinitely many replicas of the spectrum of the continuous time signal. The replicas are frequency-translated versions of the spectrum of the original signal. $\Longrightarrow$ leads to overlap iff f_s is not high enough $\Rightarrow$ ~~replicas~~ overlapping is called aliasing.

Sampling maps real frequencies in the range $[0, \infty)$ to discrete-time frequencies in the range $[0, f_s/2)$

$\downarrow$

The Nyquist frequency.

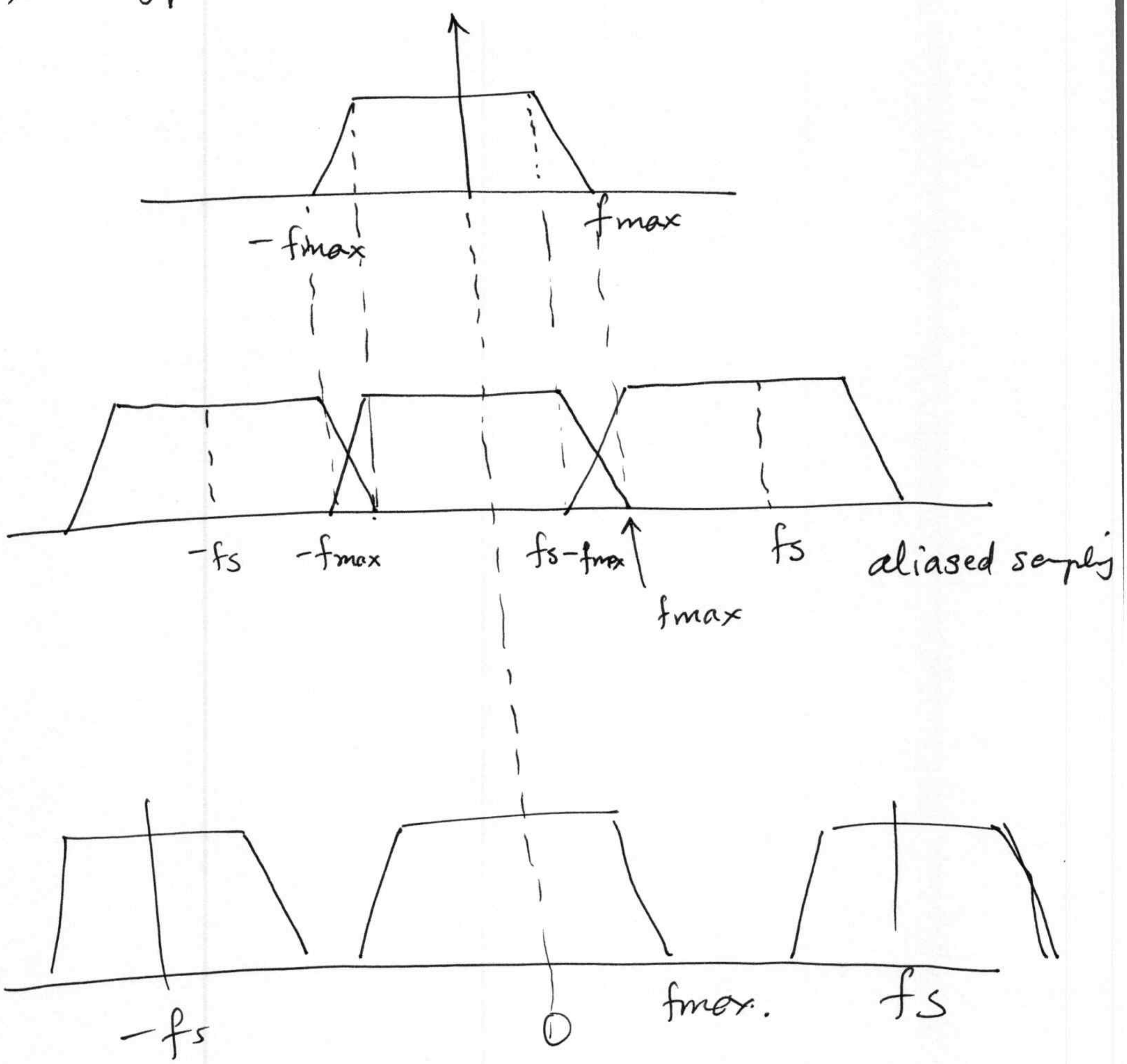
$$\frac{f_s}{2} = f_{max}$$

or $$\frac{f_s}{2} \geq f_{max}$$

or $$f_s \geq 2 f_{max}.$$

Nyquist criteria to avoid aliasing.

$2f_{max}$: Nyquist sampling frequency.



$$f_s - f_{max} > f_{\cancel{s}max}$$

$$\cancel{2f_s} > f$$

$$f_s > 2f_{max}.$$

→ Sampling a complex valued signal (For later).

→ One question on Quantization (Reading Assignment).

<u>Bandpass sampling</u>: Consider a real valued signal

centered at 70 MHz with a bandwidth of 20 MHz

How high should f_s be?

$$\underbrace{\frac{2f_s + W}{n+1}}_{\frac{2f_u}{n+1}} < f_s < \underbrace{\frac{2f_c - W}{n}}_{\frac{2f_L}{n}}$$

Highest frequence is $70 + 20/2 = 80$

So $\quad f_{\text{Nyquist sampling}} = 2 \times 80 = 160\text{ MHz}$

too high!

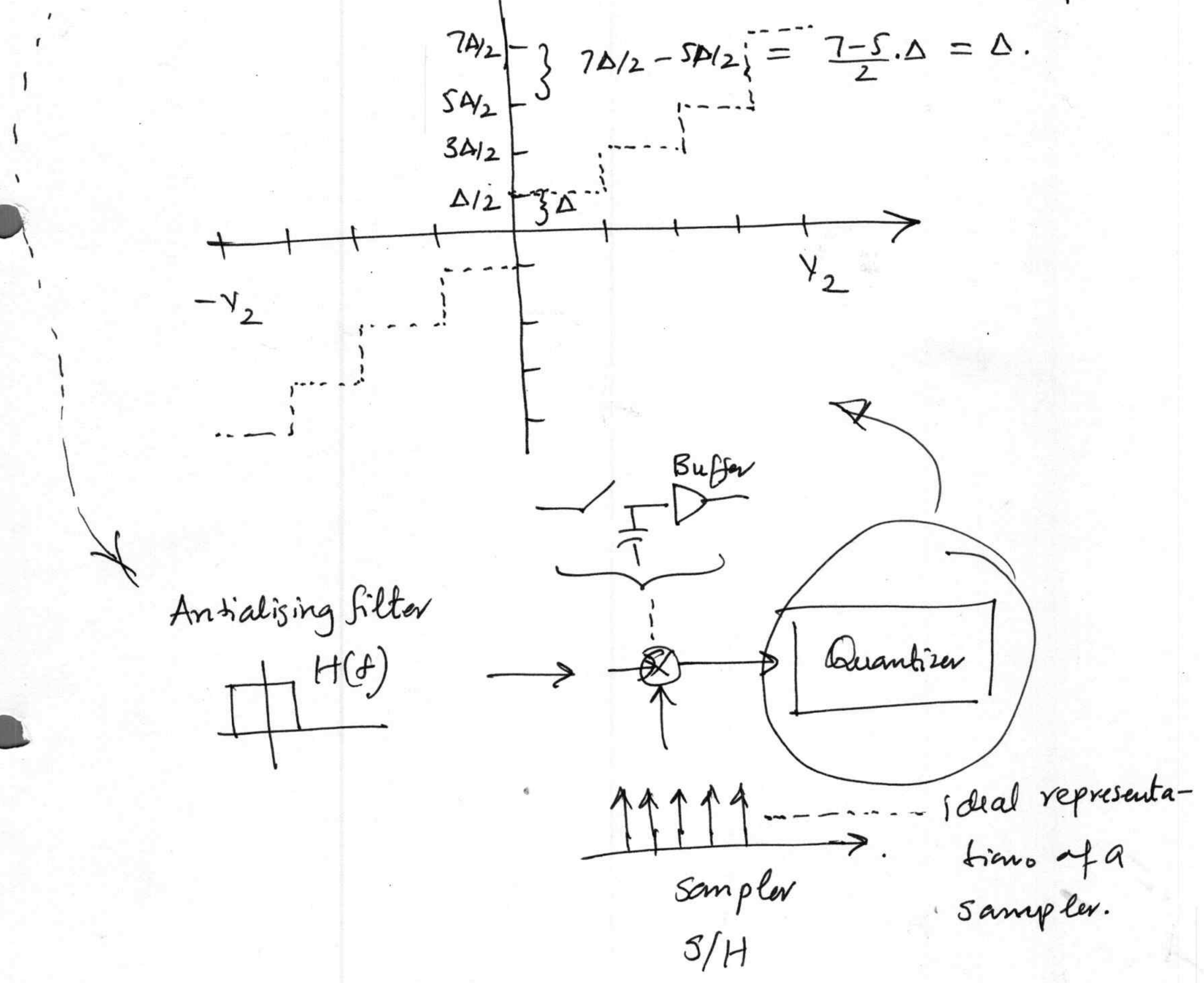
⋮

to be done.

2.7.4 Quantization

As we know from the following is the block diagram for an ADC.

Figure 2-22 : i/p - o/p relation of a symmetric, mid rise quantizer.



The quantizer converts the constant value provided by the sampler, $x[nT_s]$, into a digital number

$$x_q[n] = Q[x[nT_s]]$$

Quantizers are characterized by three main parameters:

(1) The range

(2) The number of bits

(3) The distribution of the quantization levels.

A signal outside the range of quantization will be clipped, resulting in? (Answer: Non-linear distortion).

↓

severe and unacceptable.

↓

Solution: the i/p signal is limited to the range $\left[-\frac{V}{2}, \frac{V}{2}\right]$ through scaling.

OR

$$\mathbb{P}\left(v(t) \notin [-V/2, V/2]\right) \le \epsilon \approx 0.$$

almost.

Range of i/p ~~range~~ voltage of the quantizer is always specified (1 V pk-pk to 2 V pk-pk).

The number bits, b.

Quantization levels: 2^b. (May or may not be uniform).

Fig 2-22 shows a uniform quantizer that is symmetric around zero ⟶ called <u>mid-rise</u> quantizer.

↓

o/p can never be zero.

Question: which integer/rational value a mid-rise quantizer can never take between $-\frac{\Delta}{2}$ and $\frac{\Delta}{2}$.

<u>Question:</u> If we make one level to be zero the quantizer will not be symmetric (why?)

<u>Answer:</u> Even number of levels.

<u>Quantization error:</u> Quantizer introduces error.

Quantization values are either rounded up or rounded down.

Error can be modelled as:

$$x_q[n] = x[n] + e[n] \qquad \text{(uncorrelatedness)}$$

R.V. → Question: is this true

1. $x[n]$ is uncorrelated with $e[n]$.
2. clipping is avoided.

} Model assumptions.

3. For unifor quantization: step size is equal to the least significant bit, of the quantized signal.

$$= \frac{V}{2^b} = \Delta$$

or $\quad V = 2^b \Delta$.

Under 1 + 2 + 3 the quantization error has a uniform density on $[-\Delta/2, \Delta/2]$

$$f(x) = \begin{cases} 1/\Delta, & -\Delta/2 \le x \le \Delta/2 \\ 0 & \text{else.} \end{cases}$$

Thus, $\quad E\left[e[n]\right] = 0$.

and $\quad \sigma_q^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} x^2\,dx = \frac{\Delta^2}{12} = \frac{V^2}{12\left(2^{2b}\right)}$

Question: how can we reduce the power in quantization error?

Answer: $\sigma_q^2 \xrightarrow[b \to \infty]{} 0$

Rather σ_q^2 a more useful metric is signal-to-quantization noise ratio.

$$SQNR = \frac{P_{avg}}{\sigma_q^2}$$

Define the peak-to-average power ratio, (PAPR) as

$$\eta = \frac{P_{peak}}{P_{avg}}$$

$P_{peak} = \max |x(t)|^2$ (instantaneous power).

out m (20/10/2025)
→ HW 3 p1. (a) Calculate the SQNR of the signal:

$$A\cos(2\pi f_0 t + \theta_1) + A\sin(2\pi f_0 t + \theta_2)$$

~~quanti~~ quantized to b bits.

HW3, p1 (b) Obtain a numerical value if the parameters are

$A = 1V$

$f_0 = 60Hz$.

$\theta_1 = \frac{\pi}{2}$ rads

$\theta_2 = \frac{3\pi}{2}$ rads.

$b = 8$ bit quantizer. (uniform and midrise).

HW3, p2 (a) + (b) Repeat problem 1 for a Gaussian distribu random signal ~~pulse~~ with $\sigma = 1$ and the quantizer limited to limited to $-\frac{3}{2}V$ to $\frac{3}{2}V$.

2-7.5: Pulse-code modulation:

- Misnomer, not modulation.
- ADC method.
- Introduced in the 70's
- Denotes sampling of toll-quality signal @ ~~8kb/s~~ 8ksamples/s (8000 baud rate) and quantization to 8-bits per sample.
- Narrowband speech 300Hz — 3.4kHz.

Prior to sampling an anti-aliasing filter is used to limit the freq. range to 300 Hz – 3.4 kHz.

- Voice signal is encoded into 64 kb/sec digital data stream for transmission.
- The relevant ITU standard is G.711.

---

quiz. Sampling any signal of bandwidth W using a sampling frequency $f_s > 2W$, ~~following~~ followed by quantization with b-bits of resolution to achieve a data rate of 2bW, can be called PCM.

---

quiz. – SQNR is considered to be the quality of quantized signal.

quize. – Reading assignment / Quiz: Non uniform quantization.

quiz – Speech signals do not fall into category of uniformly distributed signal in the quantization range.

quiz. – Mention in class: read non-uniform quantization.

- SQNR, for uniformly distributed signals, is proportional signal power.
- perceptually this means that low voltage speech parts will experience low quality — not desirable.

Chapter 3: Communication Channels

- Understand the channel + its attributes is the FIRST task undertaken by the comm systems design team.
- Mathematical properties are needed to tailor the the design to the channel.

quiz. — Concepts:—

1. Bandpass and lowpass transmission.
2. Noise
3. Attenuation
4. Capacity

Definition: (Channel) Everything between a t_x and r_x.

Examples:

(a) Wireless

(b) Optical Fibre

(c) Metallic (copper/aluminum) wire

(d) Co-axial cable.

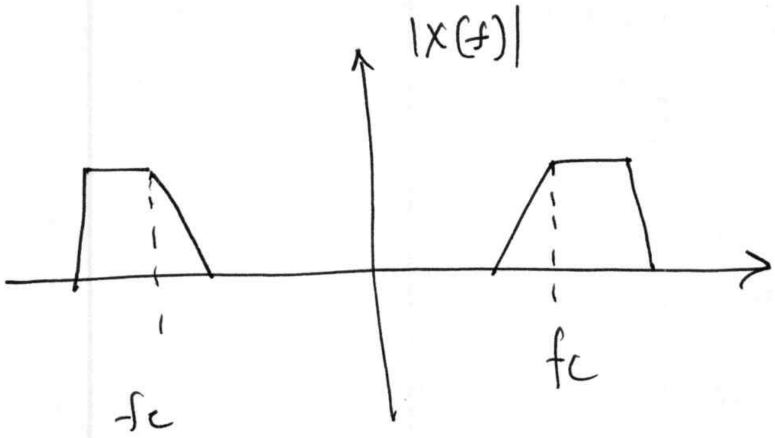
- quiz.
  Channel constraints: Physical channels have constraints.

  - frequency band
  - power

  → on the type signals that can be transmitted.

- quiz.
  Physical channels are raw: These properties cannot be controlled by a communication system. → quiz. explains why systems have to be tailored to the channel and not the other way around. And why we need to study channels in the first place.

3-2: Bandpass and Low-pass Channels



quiz-

- Most signals in comm. systems are real and <u>bandpass</u>

- Example the current flowing through an antenna is a real-valued signal, and therefore the emitted (RF) signal after modulation is real-valued + bandpass. example is signal received in a car radio.

- bandpass signal / 2 sided bandwidth 2W centered on fc; $f_c \gg 2W$ ($f_c > 2W$),

- symmetric around $f=0$

- But not necessarily around f_c.

quiz.

Every real-valued bandpass signal has one-tone relationship with a complex lowpass (or baseband) signal.

— Generally the term band-limited is used synonomously with "low-pass equivalent".

$$\underset{\text{base-band}}{x_{bb}(t)} \longleftrightarrow \underset{\text{bandpass}}{x(t)}$$

Some FT and time domain properties:

$$x^*(t) \overset{\mathcal{F}}{\Longleftrightarrow} \underline{X}^*(-f)$$

The FT of the cojugate is the cojugate of frequency-reversed Fourier transform.

HW3 : If $x(t)$ is real-valued, its FT is cojugate symmetric.

$$\underline{X}(f) = \underline{X}^*(-f)$$

$$\Downarrow$$

Magnitude is symmetric (an even function):

$$|\underline{X}(f)| = |\underline{X}(-f)|.$$

the phase is anti-symmetric.

$$\arg \underline{X}(f) = -\arg \underline{X}(-f)$$

HW3 prove this. (without using integrals).

- HW3: prove the converse: If the FT is cojugate symmetric then the signal $x(t)$ is real-valued.

$\left\{ \text{Representation of a real-valued bandpass signal by a complex-valued baseband signal.} \right.$

- Analogous to representation of a ~~comple~~ sinusoid with a complex valued phasor.

(1) The frequency of the sinusoid does not appear in the phasor $\longleftrightarrow$ as the center frequency of the bandpass signal f_c does not appear in the baseband signal.

To derive the 1-1 relationship between $x_{bb}(t) \longleftrightarrow x(t)$

shift the one-sided spectrum of $x(t)$ to the left by f_c,

and multiply by two:→.

$$\underline{X}_{bb}(f) = \begin{cases} 2\,\underline{X}(f+f_c) & f > -f_c \\ 0 & \text{else}. \end{cases}$$

- $\underline{X}_{bb}(f)$ is not necessarily symmetric with respect to $f=0$. ⇒ it is the Fourier transform of some yet to be determined, complex-valued signal:

$$x_{bb}(t) = a(t)\,e^{j\phi(t)} = a(t)\left\{\cos[\phi(t)] + j\sin[\phi(t)]\right\}$$

$$= \underbrace{a(t)\cos[\phi(t)]}_{\substack{\text{real part} \\ = Re\{x_{bb}(t)\} \\ \text{in-phase} \\ \text{component} \\ I}} + j\,\underbrace{a(t)\times\sin[\phi(t)]}_{\substack{\text{imaginary} \\ \text{part} \\ Im\{x_{bb}(t)\} \\ \text{quadrature} \\ \text{component} \\ Q}}$$

-

$$x_{bb}(t) = x_I(t) + j\,x_Q(t).$$

$$x_I(t) = \frac{1}{2}\{x_{bb}(t) + x_{bb}^*(t)\}$$

$$x_Q(t) = \frac{1}{2j}\{x_{bb}(t) - x_{bb}^*(t)\}$$

Why complex baseband? $\rightarrow$ because it is independent of f_c. $\longrightarrow$ As a result it sometimes necessary to go from real-valued bandpass to complex-valued baseband. (or lowpass equivalent).

Now:

$$x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} \underline{X}^*(-f)$$

we have:

$$\underline{X}_I(f) \longleftrightarrow \frac{1}{2}\{\underline{X}_{bb}(f) + \underline{X}_{bb}^*(-f)\}$$

and

$$\underline{X}_Q(f) \longleftrightarrow \frac{1}{2j}\{\underline{X}_{bb}(f) - \underline{X}_{bb}^*(-f)\}.$$

(Please note: $\underline{X}_I(f) \neq Re\{\underline{X}_{bb}(f)\}$

and $\underline{X}_Q(f) \neq Im\{\underline{X}_{bb}(f)\}$

even though the relations hold true in time domain.) AND ~~also~~ also, $\underline{X}_{bb}(f) = \underline{X}_I(f) + j\underline{X}_Q(f)$

So how do we go from $\underline{X}_{bb}(f)$ to $\underline{X}(f)$.

"Shift $\underline{X}_{bb}(f)$ to the right, shift the cojugated frequency reversed $\underline{X}_{bb}^*(-f)$ to the left, scale by $\frac{1}{2}$."

HW3

$$\underline{X}(f) = \frac{1}{2}\left\{ \underline{X}_{bb}(f-f_c) + \underline{X}_{bb}^*(f-f_c)\right\} \qquad (3.11)$$

To determine the time-domain relationship between $x(t)$ and $x_{bb}(t)$, we

$$x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} \underline{X}^*(-f)$$

$$\Re e\{x(t)\} = \frac{x(t) + x^*(t)}{2} \longleftrightarrow \frac{X(f) + \underline{X}^*(-f)}{2}$$

$$\Im m\{x(t)\} = \frac{x(t) - x^*(t)}{2} \longleftrightarrow \frac{X(f) - \underline{X}^*(f)}{2}$$

(3.11) suggests that:

$$x(t) = \frac{1}{2} x_{bb}(t) e^{j2\pi f_c t} + \frac{1}{2} x_{bb}^*(t) e^{-j2\pi f_c t}$$

$$= \frac{1}{2}\left\{ \underline{X}_I(t) + j\,\underline{X}_Q(t)\right\} e^{j2\pi f_c t} + \frac{1}{2}\left\{ \underline{X}_I(t) - j\,\underline{X}_Q(t) \right\} e^{j2\pi f_c t}$$

$$= \frac{1}{2}\left\{x_I(t) + j x_Q(t)\right\}\left(\cos(2\pi f_c t) + j\sin(2\pi f_c t)\right\}$$

$$+ \frac{1}{2}\left\{x_I(t) - j x_Q(t)\right\}\left(\cos(2\pi f_c t) - j\sin(2\pi f_c t)\right\} \quad (3.13)$$

HW3

multiply this out to find that $x(t)$ is the real part of the complex multiplication:

$$x(t) = \Re e\left\{\left[x_I(t) + j x_Q(t)\right]\left[\cos(2\pi f_c t) + j\sin(2\pi f_c t)\right]\right\}$$

$$= \Re e\left\{x_{bb}(t) e^{j2\pi f_c t}\right\} \quad \text{———} \quad (3.14)$$

Equation (3.14) suggests that a real-valued bandpass signal can be obtained from its complex baseband $x_{bb}(t)$ by multiplication with the complex exponential exponential $e^{2j\pi f_c t}$, followed by discarding the imaginary part, Note the similarity to obtaining a sinusoidal signal from its phasor. If the operations from (3.13) are carried out, we obtain the following representation:

$$x(t) = \left[x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)\right] \quad (3.16)$$ (17)

(passband signal from inphase and quadrature

phase)

components.)

It is possible to avoid complex arithematic entirely. Eq. (3.14) and (3.16) are the foundation of up and down conversion.

— The most property of x_I and x_Q is that:

$$\langle x_I, x_Q \rangle = 0 \quad \text{or} \quad x_I(t) \perp x_Q(t)$$

HW3 prove it ?

$$\langle x_I(t)\cos(2\pi f_c t), x_Q(t)\sin(2\pi f_c t)\rangle = 0$$

Following Parsevals we can take this inner product into the frequency domain.

$$x_I(t)\cos(2\pi f_c t) \longleftrightarrow \frac{1}{2}\left\{\underline{X}_I(f-f_c) + \underline{X}_I(f+f_c)\right\}$$

(3.18a)

and

$$x_Q(t)\sin(2\pi f_c t) \longleftrightarrow \frac{1}{2j}\left\{\underline{X}_Q(f-f_c) - \underline{X}_Q(+f+f_c)\right\}$$

- and Inner product is equivalent to:

<u>HW3</u>:

$$\frac{1}{4}\int_{-\infty}^{+\infty}\left[\underline{X}_I(f-f_c) + \underline{X}_I(f+f_c)\right]\cdot j\left[\underline{X}_Q^*(f-f_c) - \underline{X}_Q^*(f+f_c)\right] df$$

why?

If we open the brackets, there will be four products,

- Two of the products

$$\underline{X}_I(f-f_c)\,\underline{X}_Q^*(f+f_c) = 0$$

and $$\underline{X}_I(f+f_c)\,\underline{X}_Q^*(f-f_c) = 0$$

why? HW#3.

are equal to zero as long as the carrier frequency is greater than the bandwidth i.e., $f_c > W$.

The sum of the other two products is zero:

$$\int_{-\infty}^{+\infty} \underline{X}_I(f-f_c)\,\underline{X}_Q^*(f-f_c)\,df - \int_{-\infty}^{+\infty} \underline{X}_I(f+f_c)\,\underline{X}_Q^*(f+f_c)$$

$$=0.$$

Why do we care about $\langle \underline{X}_I, x_Q \rangle = 0$?

- Because it allows us to modulate the inphase and quadrature components independently.

The orthogonality of $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ suggests a geometric anology. As shown in Fig 3-5, $x_I(t)$ is the projection onto the axis defined by $\cos(2\pi f_c t)$, and $x_Q(t)$ is the projection onto the axis defined by $-\sin(2\pi f_c t)$. This is why x_I is referred to as the in-phase components and x_Q is referred to as the quadrature component.

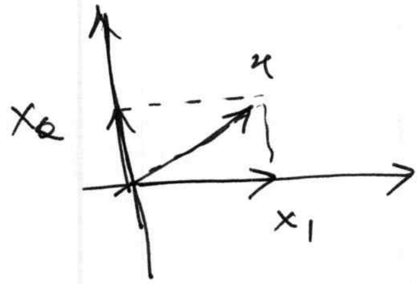


Figure 3-5: The baseband components are the ~~verification~~ projections of the passband signal onto the axes the axes defined by $\cos(2\pi f_c t)$ and $-\sin(2\pi f_c t)$.

We have:

$$x(t) = \Re e\{[a(t)e^{j\phi(t)}]e^{2j\pi f_c t}\}$$

$$= \underline{a(t)} \cos[2\pi f_c t + \phi(t)]$$

natural envelope of the signal $x(t)$

$x_{bb}(t)$: complex envelope

~~at~~ $a(t) = \sqrt{x_I^2(t) + x_Q^2(t)} = |x_{bb}(t)|$

So far we have gone from $x_{bb}(t)$ to $x(t)$.

What about $x(t) \longrightarrow x_{bb}(t)$ [the reverse direction].

For that we need to se Eq (3.4) in time domain:

$$\underline{X}_{bb}(f) = \begin{cases} 2\underline{X}(f+f_c) & f > -f_c \\ 0 & \text{otherwise} \end{cases}$$

Define

$$\underline{X}_+(f) = \begin{cases} X(f) & f > 0 \\ 0 & \text{else} \end{cases}$$

$$\underline{X}_-(f) = \begin{cases} 0, & f > 0 \\ \underline{X}(f) & \text{otherwise} \end{cases}$$

$\underline{X}_+(f)$ and $\underline{X}_-(f)$ are one-sided spectra

We can compress this relationships can be written as:

$$\underline{X}_+(f) = \frac{1 + \text{sign}(f)}{2}\underline{X}(f) = \frac{1}{2}\underline{X}(f) + \frac{\text{sgn}(f)}{2}\underline{X}(f)$$

$$\underline{X}_-(f) = \frac{1 - \text{sgn}(f)}{2}\underline{X}(f) = \frac{1}{2}\underline{X}(f) - \frac{\text{sgn}(f)}{2}\underline{X}(f)$$

$$\text{sgn}(f) = \begin{cases} 1, & f \geq 0 \\ -1, & f < 0. \end{cases}$$

(27/10/2025)

[Monday] ADC: Channels (Continued from previous lectures).

DEFINITION: A complex-valued signal with one-sided-spectrum is called ANALYTIC. → an analytic signal has not frequency component for $f<0$. ⇒ $x_+(t)$ and $x_-(t)$ are both analytic.
(or $f>0$)

With these notations, Eq (3.4),

$$\underline{X}_{bb}(f) = \begin{cases} 2\underline{X}(f+f_c) & f > -f_c \\ 0 & \text{else} \end{cases} \qquad (3.4)$$

can be written as:

$$\underline{X}_{bb}(f) = 2\underline{X}_+(f+f_c)$$

or in time-domain:

$$x_{bb}(t) = 2x_+(t)\, e^{-j2\pi f_c t}$$

Q: In turn, how can we obtain $x_+(t)$ from $x(t)$?

Ans. $x_+(t)$ and $x_-(t)$ are not only complex-valued but complex-conjugate of each other, because $x_+(t) + x_-(t) = x(t)$

$$\cancel{x_+^*(t) = x^*(t) - x_-^*(t)}$$

$$\cancel{= x(t) - x_-^*(t)}$$

quiz.

$$x_+(t) + x_-(t) = x(t) = x_+^*(t) + x_-^*(t)$$

$$\underset{\substack{x^*(t) \\ \overline{x(t) \text{ is real bandpass.}}}}{\parallel}$$

$$x_+(t) = \frac{x_{bb}(t)\, e^{j2\pi f_c t}}{2}$$

$$x_+^*(t) = \frac{x_{bb}^*(t)\, e^{-j2\pi f_c t}}{2}$$

$$x_+(t) - x_+^*(t) = \frac{1}{2}\left\{x_{bb} \cancel{-} x_{bb}^*(t)\right\} e^{j2\pi f_c t} - e^{-j2\pi f_c t}$$

$$\Im m\{x_+(t)\} + \Im m\{x_-(t)\} = \Im m\{x(t)\}$$

$$= 0.$$

$$\Im m\{x_+(t)\} = -\Im m\{x_-(t)\}.$$

quiz

Also, $f(t)$ real $\rightarrow$ $\mathrm{Re}\,F(\omega)$ is even.

Inverse transform.

$$\mathrm{Re}\{\underline{X}(f)\} \overset{\mathcal{F}}{\longleftrightarrow} \frac{x(t)+x(-t)}{2}.$$

so $$\mathrm{Re}\{\underline{X}_+(f)\} \overset{\mathcal{F}}{\longleftrightarrow} \frac{x_+(t)+x_+(-t)}{2}$$

$$\mathrm{Re}\left\{\frac{(1+\mathrm{sgn}(f))}{2}\underline{X}(f)\right\} \overset{\mathcal{F}}{\longleftrightarrow} \cancel{\mathrm{Re}\{}$$

$$\cancel{\frac{1}{2}\left\{\left(1+\frac{1}{jt\pi}\right) * x(t)\right.}$$

$$\cancel{\left.+\left(1-\frac{1}{jt\pi}\right) * x^*(t)\right\}}.$$

$$\cancel{\frac{1}{2}\left\{\left(1+\frac{1}{jt\pi}\right) *\right.}$$

$$\mathrm{Re}\Big\{ \underbrace{\frac{1+\mathrm{sgn}(f)}{2} X(f)}_{X_+(f)} \Big\} \xleftrightarrow{\mathcal{F}} ?$$

$$= \frac{1}{2}\left[\left(\frac{1}{2}\delta(t) + \frac{1}{2}\frac{1}{j\pi t}\right) * x(t) + \left(\frac{1}{2}\delta(-t) - \frac{1}{j\pi t}\right) * x(-t\right.$$

$$= \frac{1}{2}\left[\left(\delta(t) + \frac{1}{2}\frac{1}{j\pi t} - \frac{1}{2}\frac{1}{j\pi t}\right)\right] * x(t)$$ (Assuming $x(t)$ is even.)

$$= \frac{1}{2}\left[\delta(t) * x(t)\right] = \frac{x(t)}{2}$$

$$\mathrm{Re}\{x_+(t)\} + \mathrm{Re}\{x_-(t)\} = \mathrm{Re}\{x(t)\}.$$

$$\frac{x(t)}{2} + \mathrm{Re}\{x_-(t)\} = x(t)$$

$$\mathrm{Re}\{x_-(t)\} = \frac{x(t)}{2}$$

Similarly $\mathrm{Re}\{x_+(t)\} = \dfrac{x(t)}{2}$

Thus, $\mathrm{Re}\{x_+(t)\} = \mathrm{Re}\{x_-(t)\}$.

So,
$$x_+^* = x_-$$
$$x_-^* = x_+.$$

Thus, we can write

$$x_+(t) = \frac{x(t)}{2} + j\frac{\hat{x}(t)}{2}$$
$$= \frac{1}{2}\left\{x(t) + j\hat{x}(t)\right\}$$
$$x_-(t) = \frac{1}{2}\left\{x(t) - j\hat{x}(t)\right\}$$

The signal $\hat{x}(t)$ is called the <u>Hilbert transform</u> of $x(t) \longrightarrow$ next section.

Now,
$$x_{bb}(t) = 2x_+(t)\,e^{-j2\pi f_c t}$$
$$= 2\cdot\frac{1}{2}\left\{x(t) + j\hat{x}(t)\right\}e^{-j2\pi f_c t}$$
$$= \left\{x(t) + j\hat{x}(t)\right\}e^{-j2\pi f_c t}$$

$$= x(t)\cos(2\pi f_c t) + \hat{x}(t)\sin(2\pi f_c t)$$

$$+ j\left[-x(t)\sin(2\pi f_c t) + \hat{x}(t)\cos 2\pi f_c t\right]. \quad -(3.29)$$

Thus the in-phase and quadraturature phase

signals are:

(3.30a) (3.30b)

$$\begin{cases} x_I(t) = x(t)\cos(2\pi f_c t) + \hat{x}(t)\sin(2\pi f_c t) \\ x_Q(t) = -x(t)\sin(2\pi f_c t) + \hat{x}(t)\cos(2\pi f_c t) \end{cases}$$

(3.29) involves complex arithmetic. — inconvinient in digital hardware — we use $\cos(2\pi f_c t) \perp \sin(2\pi f_c t)$ to avoid this.

Consider (3.16):

$$x(t) = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)$$

$$\Rightarrow x(t)\cos(2\pi f_c t) = x_I(t)\cos^2(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)\cos(2\pi f$$

$$=$$

$$2\cos^2(2\pi f_c t) = \cos(4\pi f_c t) + 1$$

$$2\cos(2\pi f_c t)\sin(2\pi f_c t) = \sin(4\pi f_c t)$$

The components at twice the carrier frequency can be filtered out.

- $$\text{LPF}\left[x_I(t)\, 2\cos(2\pi f_c t)\right] = x_I(t)$$

which in frequency domain means that

$$\text{LPF}\left[\underline{X}(f-f_c) + \underline{X}(f+f_c)\right] = \underline{X}_I(f)$$

- Since we require the BW of the bandpass signal to be $2W \ll f_c$, the only requirement is towards having an LPF that has a cutoff frequency between ~~f~~ W and $(f_c - W)$

In a similar fashim we can recover x_Q by

$$-\mathrm{LPF}\left[x(t)\sin(2\pi f_c t)\right] = x_Q(t)$$

and $$j\,\mathrm{LPF}\left[X(f-f_c) - X(f+f_c)\right] = X_Q(f)$$

---

(1) The baseband signal $x_{bb}(t)$ is independent of of the carrier frequency.

$\Downarrow$

Implementing filtering @ baseband is much ~~more~~ less computationally demanding.

SECTION 3-3 Hilbert Transform

$\hat{x}(t)$

$\uparrow$ Hilbert transform.
(not a time-freq. transformation)

time-domain signal $\xrightarrow{\text{Hilbert transform}}$ time-domain signal

The device that performs the Hibert trans-form ~~can~~ is called a Hibert Transformer — can be either analog or digital. Consider:

(3.24a) $$X_+(f) = \frac{1 + \text{sgn}(f)}{2} \cdot X(f)$$

(3.24b) $$X_-(f) = \frac{1 - \text{sgn}(f)}{2} \cdot X(f)$$

(3.28a) $$x_+(t) = \frac{1}{2}\left[x(t) + j\hat{x}(t)\right].$$

(3.28b) $$x_-(t) = \frac{1}{2} x(t) - j\hat{x}(t)$$

Comparing 3.24a ↔ 3.28a.

$$\cancel{x_+(t) = \frac{x(t)}{2} + \frac{\text{sgn}(f)}{2}}$$

$$X_+(f) = \frac{X(f)}{2} + j\frac{\hat{X}(f)}{2}$$

$$\frac{X(f)}{2} + \frac{\text{sgn}(f)}{2} X(f) = j\frac{\hat{X}(f)}{2} + \frac{X(f)}{2}.$$

$$\boxed{-j\,\text{sgn}\,X(f) = \hat{X}(f)}$$

$$\hat{x}(t) \xleftrightarrow{\mathcal{F}} -j\,\mathrm{sgn}(f)\,X(f)$$

This determines the FT of $\hat{x}(t)$.

Thus in the freq. domain a Hilbert transformer is an LTI system with frequency response:

$$\cancel{H(f) = -j\,\mathrm{sgn}(f)\,N}$$

$$H(f) = \frac{Y(f)}{X(f)}$$

$$= \frac{-j\,\mathrm{sgn}(f)\,X(f)}{X(f)}$$

$$H(f) = -j\,\mathrm{sgn}(f)$$

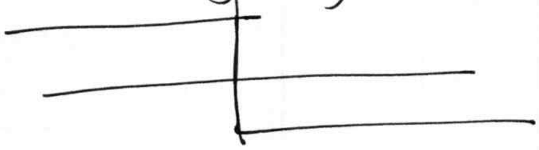
$$\boxed{\text{An ideal Hilbert transforms increases the phase of the signal by } 90^\circ\ (\pi/2 \text{ rad}) \text{ degrees for negative freq. and decrease by } 90^\circ \text{ for postive freq.}}$$

— A~~n~~ Hilbert transformer is type of an all-pass filter because $|H(f)| = 1$.

— Phase response is discontinuous. Sharp trans~~f~~ition from 90° to −90° at $f=0$.

$$\arg H(f) = \begin{cases} \pi/2 & f<0 \\ -\pi/2 & f>0 \end{cases}$$

Arg H(f).



Quiz. — Real Hibert transformers can only only approximate Ideal Hilbert transformers because?

— The phase is discontinuous. at $f=0$.

Assumption: The filter is analog.

— Digital filters can indeed have discontinuous phase.

for DSP course.

- An ideal Hilbert transformer has an infinite impulse response and is non-causal.

- Even an IIR filter can only approximate a Hilbert transformer. beause

  1. HT is non-causal and of infinite duration.
  2. The phase repsne $+90^\circ$ and -90° is flat/constant over an infinite band. The IIR filter can only maintain it for finite bands.
  3. Stability is another issue. HT is not stable $\Rightarrow$ not physically realizable.

---

quiz

Examples: HT of complex exponentials:

$\rightarrow$ $e^{j2\pi f t}$

$$\mathcal{H}\left(e^{j2\pi f t}\right) = \begin{cases} e^{j2\pi f t + \pi/2} & f < 0 \\ e^{j2\pi f t - \pi/2} & f > 0 \end{cases}$$

?

As a more general example:-

Hbl3 $\longrightarrow$ $\mathcal{H}\{x(t)\cos(2\pi f_c t)\} = x(t)\sin(2\pi f_c t)$

If $x(t)$ has no component above f_c.

Let us determine the impulse response of the Hilbert transformer.

$$sgn(f) = \lim_{a \to 0}\left[e^{-af}u(f) - e^{af}u(-f)\right]$$

$$\int_0^\infty e^{-af} e^{j2\pi ft}\, df = \frac{1}{a - j2\pi t}$$

$$\int_0^\infty e^{af} e^{j2\pi ft}\, df = \frac{1}{a + j2\pi t}$$

and

$$\lim_{a \to 0}\left[\frac{1}{a - j2\pi t} - \frac{1}{a + j2\pi t}\right] \overset{\mathcal{F}}{\longleftrightarrow} sgn(f)$$

which gives

$$-\frac{1}{j\pi t} \xleftrightarrow{\mathcal{F}} sgn(f)$$

Therefore

$$\boxed{h(t) = \frac{1}{\pi t}}$$

Impulse response of Hilbert transformer.

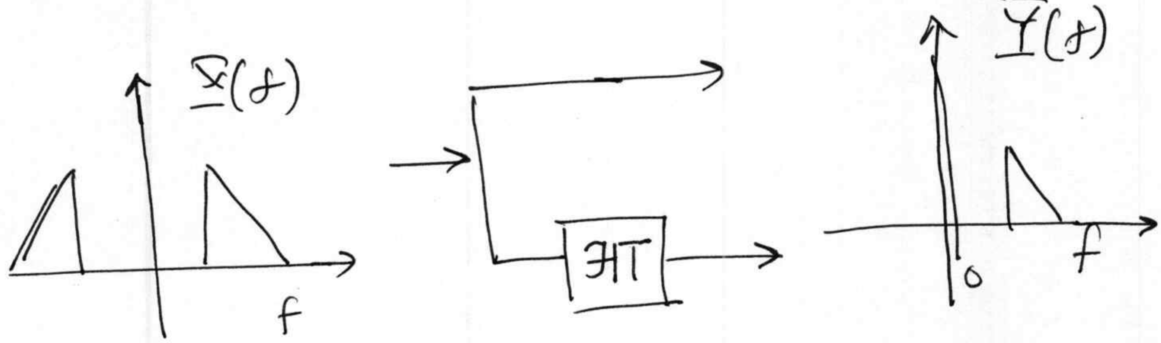
(Wednesday lecture) ADC

5/11/2025 [Not started from here]

(continued from page 27)

PROPERTIES OF THE HILBERT TRANSFORMER:

We need these properties to study Analog Signal Processing and transceiver design in Chapter4.



A phase spliter circuit → spliting a real-bandpass signal into an complex-valued bandpass signal, which is analytic.

The two real o/p signals are considered of one complex-valued signal. Thus signal is:

$$x(t) + j\hat{x}(t)$$

$$= 2x_+(t)$$

The impulse response of the phase splitter is:

$$h(t) = \delta(t) + j\frac{1}{\pi t}$$

The Hilbert transform $\mathcal{H}$ has the following properties:

- If x is real $\Rightarrow$ $\hat{x}$ is also real, Also

$$\mathcal{H}\{\hat{x}(t)\} = -x(t)$$

$\Rightarrow$ Applying the Hilbert transform twice is equivalent to multiplying by -1.

e.g. $\cos(2\pi f_c t) \xrightarrow{\mathcal{H}} \sin(2\pi f_c t) \xrightarrow{\mathcal{H}} -\cos(2\pi f_c t)$.

- $\perp$ (orthogonality) $\langle x(t), \hat{x}(t)\rangle = 0$

Proof is based on on the Parseval's Theorem:

and $\quad |X(-f)| = |X(f)| \quad$ [for real $x(t)$].

$$\langle x(t), \hat{x}(t)\rangle = \int_{-\infty}^{+\infty} x(t)\,\hat{x}(t)\,dt$$

$$= \int_{-\infty}^{+\infty} X(f)\, j\,\mathrm{sgn}(f)\, X^*(f)\, df.$$

$$= -j\int_{-\infty}^{0} |X(f)|^2 df + j\int_{0}^{\infty} |X(f)|^2 df$$

$$= 0 \qquad \left(\text{As } |X(f)| = |X(-f)|\right)$$

(from Hermitian Property of real functions.)

- <u>Preservation of Energy or Power:</u>

$$\langle x(t), x(t)\rangle = \langle \hat{x}(t), \hat{x}(t)\rangle$$

Parsevals: $\quad \|\hat{x}\|^2 = \|\hat{X}\|^2 = \int_{-\infty}^{+\infty} |j\,\mathrm{sgn}(f)\, X(f)|^2 df$

$$= \int_{-\infty}^{+\infty} |X(f)|^2 df$$

$$= \|X(f)\|^2 = \|x\|^2.$$

Thus, $\|x\|^2 = \|\hat{x}\|^2$.

- Finally if the sum of $\hat{x}(t)$ and $x(t)$ is passed through an LTI filter/system, the Hilbert pair relationship at the o/p — and the orthogonality — is retained. Thus.

if $\quad x(t) * h(t) = y(t)$

then $\quad [\,{}^{a}x(t) + {}^{b}\hat{x}(t)\,] * h(t) = a\,y(t) + b\,\hat{y}(t)$

The proof is simple:

$$H(f)\,\hat{X}(f) = \underline{-j\,\mathrm{sgn}(f)}\,H(f)\,X(f) = \hat{Y}(f)$$

which is the Hilbert transform of:

$$\mathcal{H}\{H(f)X(f)\} = -j\,\mathrm{sgn}\,H(f)X(f)$$
$$= \hat{Y}(f).$$

SECTION 3-4 Relationships between the Inner Products of Bandpass Signals and their Baseband Equivalent

HW3 Prove using the technology developed so far for baseband, bandpass and Hilbert transform relations that:

$$\|x(t)\|^2 = \frac{1}{2}\|x_{bb}(t)\|^2 .$$

$$= \frac{1}{2}\left\{ \|x_I(t)\|^2 + \|x_Q(t)\|^2 \right\}$$

"The energy and power of the baseband signal is twice of that of the bandpass signal".

3-5 Random Processes and Gaussian Noise:

MW 221.

- All comm systems are impacted by noise.

- Random Variables: $X : \Omega \rightarrow \mathbb{R}$.

$X(\omega) \in \mathbb{R}$.

s.t. $\{\omega : X(\omega) \leq a\} = E$

is an event

OR $\mathbb{P}(E)$ is defined $\forall a \in \mathbb{R}$.

Example: $X : \underbrace{\{HH, TH, TT, HT\}}_{\Omega} \longrightarrow$ # of heads.

$$\text{Range}\{X(\omega)\} \longrightarrow (0, 1, 2)$$

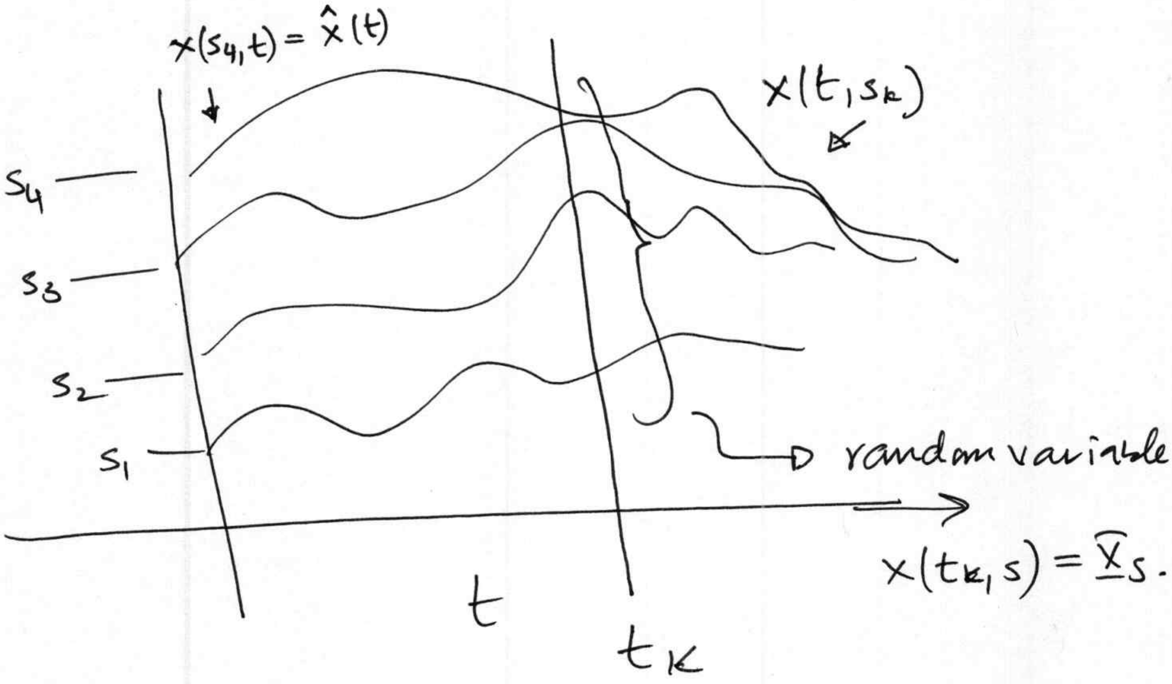
A random process is a ~~collective~~ collection (ensemble) of deterministic sample functions or signals. For each experimental outcome, there exists a deterministic function, called a sample function or a realization.

A random process can be considered as a function of two ~~random~~ variables.

$$X(s,t): \quad s \in S \qquad t \in \mathbb{R} : \text{time}.$$

sample space

$$X : S \times \mathbb{R} \rightarrow \mathbb{R}.$$



@ $t = t_k$, $\underline{X}(s, t_k) = \underline{X}_k$ is an R.V.

@ $s = s_k$, $\underline{X}(s_k, t) = x_k(t)$ is a sample function.

Random Processes can be discrete-time or continuous time.

If $\underline{X}_k$ is descrite time then the realization is x_k.

If $\underline{X}(t)$ is continuous time then the realization is $x(t)$.

W.S.S. Wide (or weak sense) stationarity.

WSS if ① mean = constant.

② autocorrelation does not change because of time shifts.

$$\mathbb{E}[\underline{X}(t)] = K.$$

$$R_{\underline{X}}(t_1, t_2) = \mathbb{E}[\underline{X}(t_1)\underline{X}^*(t_2)]$$

$$\cancel{= \mathbb{E}[\underline{X}(t_1+t)\underline{X}_2^*(t_2+t)]}.$$

$$= \mathbb{E}[\underline{X}(t_1 + t - t_1)\underline{X}^*(t_2 + t - t_1)]$$

$$= \mathbb{E}[\underline{X}(t)\underline{X}^*(t + \underbrace{t_2 - t_1}_{-\tau})]$$

$$R_{\underline{X}}(t_1, t_2) = \mathbb{E}[\underline{X}(t)\underline{X}^*(t - \tau)].$$

independent of t.

only depends on $\tau = t_1 - t_2$

Thus, $R_{\underline{X}}(t_1, t_2) = R_{\underline{X}}(t_1 - t_2) = R_{\underline{X}}(\tau)$.

Thus power is independent of time:

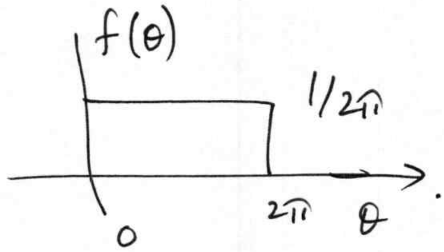
$$\|\underline{X}\|^2 = E\left[\underline{X}^2(t)\right] = R_{\underline{X}}(0) = \sigma^2 .$$

EXAMPLE 3-1: WSS Random Process:

Determine if the following random process is WSS:

$$\underline{X}(t, s) = A \cos\left[2\pi f_c t + \theta(s)\right], \text{ where } \theta \text{ is}$$

an R.V. distributed as.



Solution: The average value is:

$$\mathbb{E}\left[\underline{X}(t)\right] = \int_0^{2\pi} x(t) f(\theta)\, d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} A \cos\left[2\pi f_c t + \theta(s)\right] d\theta$$

The autocorrelation function is

$$R_X(\tau) = \mathbb{E}\left[A^2 \cos(2\pi f_c t + \theta) \cos\left[2\pi f_c (t+\tau)\right] + \theta \right]$$

$$= \mathbb{E}\left[A^2/2 \cos\left(4\pi f_c t + 2\pi f_c \tau + 2\theta\right)\right]$$

$$+ \mathbb{E}\left[A^2/2 \cos\left(2\pi f_c \tau\right)\right]$$

$$= \frac{A^2}{2} \cos(2\pi f_c \tau).$$

Thus W.S.S.

<u>Ergodicity</u>: A WSS Random process is ergodic if all statistics can be obtained from a single sample function $x(t)$, of the process.

$$\left.\begin{array}{l}\text{If a random process is} \\ \text{ergodic}\end{array}\right\} \Longrightarrow \text{Strong Sense Stationary.}$$

Ergodic Process:

Statistical Averages $\equiv$ Time Averages.

$$\mathbb{E}[x(t)] = \int_{-\infty}^{+\infty} x f(x) dx = \lim_{T\to\infty} \frac{1}{T}\int_T x(t)dt$$

$$= K \text{ (constant).}$$

Autocorrelation function is eqrual to its time autocorrelation function.

$$R_{\underline{X}}(\tau) = \mathbb{E}[\underline{X}(t)\underline{X}^*(t-\tau)] = \lim_{T\to\infty}\frac{1}{T}\int_T x(t)x^*(t-\tau)dt.$$

- Even through the Tx ~~sends~~ relays to Rx the intended signal, noise gets mixed in (added, multiplied etc.)
  - e.g. thermal noise
  - noise generated by the medium between Tx and Rx.

Consider fig ~~4~~3.7

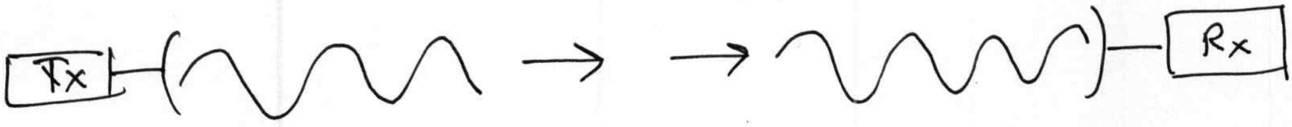
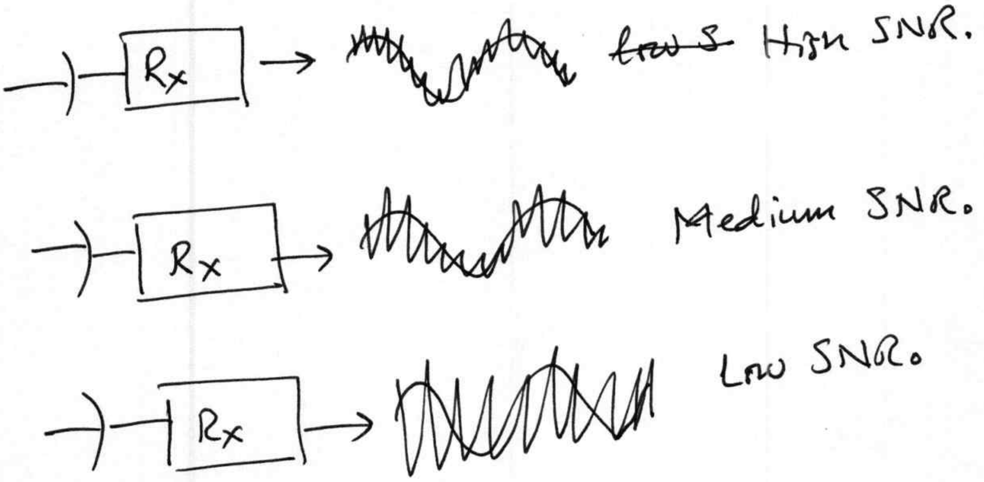


Fig 3.7



The degree to which noise may compromise the integrity of the signal is depends on the power of the signal relative to the noise, and is characterized by the SNR (Signal to Noise Ratio)

$$SNR = \frac{P_s}{P_n}$$

P_s → Power in the signal

P_n → Power in the noise

Power refers to $R_x(0)$

$$P_n \models R_n(0)$$

$$P_s \models R_s(0)$$

$R_x(0)$ is the deterministic autocorrelation as both signal and noise are deemed WSS.

- Thermal noise is produced by random Brownian motion.

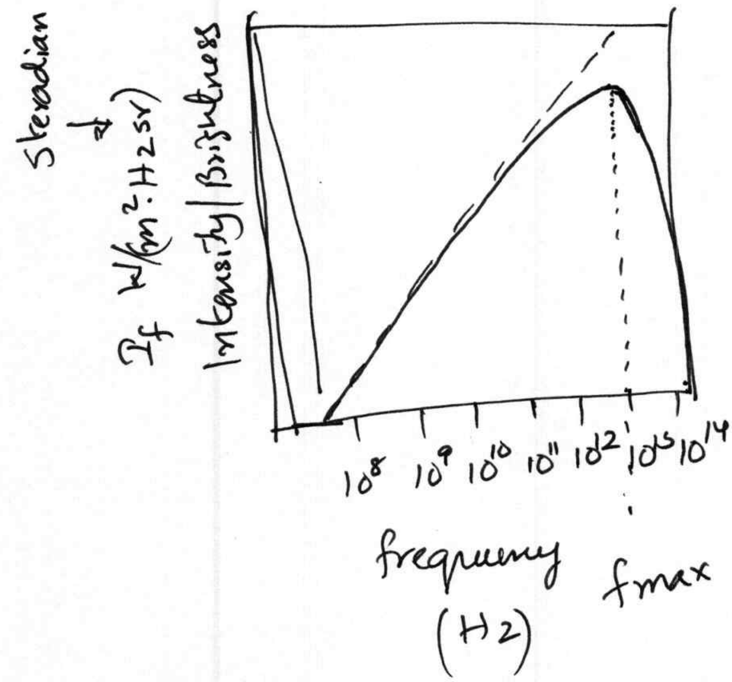
Blackbody Radiation: As noted before, thermal noise is generated by Brownian motion.

- All materials radiate and absorb EM energy.
- The amount af energy radiated + ~~em~~ absorbed by a body depends on:

  (a) T_0 (b) Shape (c) material properties

- The reference standard is called blackbody.
  - perfect absorber
  - perfect radiator

- The intensity of blackbody radiation is governed by blackbody radiation law known as Planck's blackbody radiation law given by Fig 3.8



Planck's Law: for room temperatures ($27°C = 300K$) the peak frequency radiation is at a frequency slightly higher than 10 THz (10^{13} Hz)

Rayleigh-Jean's Law: At frequencies below about 300 GHz Planck's Law can be approximated by the Rayleigh-Jean's Law.

The vast majority of comm systems work at carrier 50

frequencies below 300 GHz. (Except for the upcoming 6G systems)

which makes the Rayleigh-Jean's law perfectly applicable.

Since this is a linear relationship the power of the

thermal noise is given by:

$$P_n \propto T_c$$

$$P_n \propto W$$

T_c is called the equivalent noise temperature of the

receiver in Kelvins

W is the one-sided receiver bandwidth.

We thus have:

$$P_n = kT_cW$$
$$= N_0W$$

$\uparrow$

noise spectral density.

The receiver noise temperature T_c is related to

T_o, the physical temperature of the receiver by:

$$T_c = F T_o$$

↑

noise figure.

like a blackbody

~~F=1~~ $F = 1$ (Ideal noise free receiver)

$F > 1$ (For real, non-ideal receiver)

---

quiz.

→ Because the noise of a receiver is generated by charged particles and they are very large in numbers CLT applies. Thus any macroscopic measurement of voltage is Gaussian distributed with a PDF:

$$f(v_n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(v_n - m)^2}{2\sigma^2} \right\}$$

$$\sim N(m, \sigma^2)$$

The autocorrelation of white noise, by definition,

is:

$$R_n(\tau) = \mathbb{E}\left[v_n(t)\, v_n(t-\tau) \right] = \frac{N_0}{2}\, \delta(\tau)$$

completely uncorrela[ted]
samples at times τ
appart $\forall\, \tau > 0$.

completely independen[t]

For a WSS random process that has power signals as sample functions, we can define the Power Spectral Density as (PSD)

$$S_n(f) = \lim_{T \to \infty} \frac{1}{T}\, \mathbb{E}\left[P_n(f) \right]^2 \quad (3.73)$$

(Whenever this limit exists)

— PSD is a deterministic function $\longrightarrow$ hence very ~~itself~~ useful

- Independent of time.

- The FT of a sample function may not even exist.
  - may not be as useful for ~~ϕ~~ characterizing the signal.
  - In chapter 2 we established that the FT of the deterministic autocorrelation is the PSD.
  - According to the Weiner-Kinchine theorem(4)

(4) Einstein was the first to formulate it in his Brownian Motion paper of the existence of atoms (but he gave no proof). Also called the Weiner-Kinchine-Einstein theorem.

for WSS process $\longrightarrow$ P.T.O.

$$S_n = \cancel{R}\ \mathcal{F}\{R_n(\tau)\}$$

$$= \int_{-\infty}^{+\infty} R_n(\tau)\, e^{-j2\pi f\tau}\, d\tau$$

Given that $R_n(\tau) \; \cancel{S_n(f)} = \frac{N_0}{2}\delta(\tau)$

$$\Downarrow$$

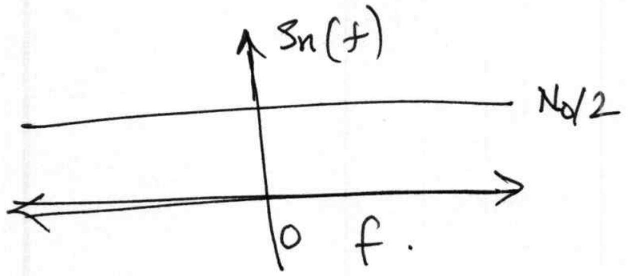
$$S_n(f) = \mathcal{F}\{R_n(\tau)\}$$

$$= \mathcal{F}\left\{\frac{N_0}{2}\delta(\tau)\right\}$$

$$= \frac{N_0}{2} \qquad -\infty < f < \infty.$$

$\Rightarrow$ Thermal noise is a WSS Gaussian process that is

white.



<u>White noise</u> uniform PSD.

$\downarrow$

power is equally ~~spa~~ distributed among all frequencies

$\hookrightarrow$ In white light all wavelengths (= frequencies) with same relative powers.

— $N_o/2$ is the two sided PSD.

— N_o is often referred to as the one-sided PSD.

Average ~~noise~~ power, PSD and autocorrelation of bandlimited noise.

$\Downarrow$

In practice, receivers are tuned to a specific frequency band and filter out the myriad of out of band signals. As a result, the white noise that impacts practical communication systems can be considered filtered by a bandpass filter, resulting in a bandlimited noise process. Therefore it is interest to investigate bandlimited noise and obtain the average power, the associated PSD and autocorrelation of bandlimited noise.

Let us assume that the input to an LTI system with a real-valued impulse response $h(t)$ is a WSS random process $\underline{X}(t)$. Then the expected value of the output is

$$\mathbb{E}[y(t)] = \mathbb{E}\left[\int_{-\infty}^{+\infty} h(\tau)\, x(t-\tau)\, d\tau\right]$$

$$= \int_{-\infty}^{+\infty} h(\tau)\, \mathbb{E}[x(t-\tau)]\, d\tau$$

$$= \mathbb{E}[x(t)] \int_{-\infty}^{+\infty} h(\tau)\, d\tau$$

$$= \mathbb{E}[x(t)]\, H(f)\big|_{f=0}$$

$$= M_{\underline{X}} H(0)$$

If the input is zero mean the output is also zero mean.

The cross correlation between input and output

is:

$$\mathbb{R}_{xy}(\tau) = \mathbb{E}\left[x(t)\, y(t-\tau)\right]$$

$$= \mathbb{E}\left[x(t) \int_{-\infty}^{+\infty} h(u)\, x(t-\tau-u)\, du\right]$$

$$= \int_{-\infty}^{+\infty} h(u)\, \mathbb{E}\left[x(t)\, x(t-\tau-u)\right] du$$

$$= \int_{-\infty}^{+\infty} h(u)\, R_x(\tau-u)\, du$$

$$= R_x(\tau) * h_u(\tau).$$

In a very similar manner it can be established

that $$\mathbb{R}_{yx}(\tau) = \mathbb{R}_x(\tau) * h(-\tau)$$

and the autocorrelation of the output is

$$\mathbb{R}_y(\tau) = \mathbb{E}\left[y(t)\, y(t-\tau)\right]$$

This gives

$$R_y(\tau) = \mathbb{E}\left[y(t)\, y(t-\tau)\right]$$

$$= \mathbb{E}\left\{ y(t) \left[h(t) * h(t-\tau)\right]\right\}$$

$$= h(\tau) * R_{yx}(\tau)$$

$$= h(\tau) * h(-\tau) * R_x(\tau)$$

So it is like this:

$$\begin{tikzcd}
R_x(\tau) \arrow[r] \arrow[d, "\mathcal{F}"] & \boxed{h(t)} \arrow[r] & \boxed{h(-t)} \arrow[r] & R_y(\tau) \\
S_x(f) \arrow[r] & \boxed{H(f)} \arrow[r] & \boxed{H^*(f)} \arrow[r] & S_y(f)
\end{tikzcd}$$

To obtain the relationship between the PSD's at the output and inputs we use the properties of the FT.

Because $h(t) \in \mathbb{R}$. $H(f)$ is conjugate symmetric. We also have

$$h(-t) \overset{\mathcal{F}}{\longleftrightarrow} H(-f) = H^*(f)$$

Therefore,

$$S_Y(f) = S_X(f)\, H(f)\, H^*(f)$$

$$= |H(f)|^2 S_X(f).$$

If we assume that input noise has white PSD $\frac{N_0}{2}$, The PSD of the output is

$$S_Y(f) = \frac{N_0}{2} |H(f)|^2.$$

To be continued.

Suppose that the filter $H(f)$ is an ideal bandpass filter with passband from $f_c - W/2$ to $f_c + W/2$:

$$H(f) = |H(f)|^2$$

$$= \begin{cases} 1 & f_c - W/2 < f < f_c + W/2 \\ 0 & \text{otherwise} \end{cases}$$

Thus, according to (8.80)

$$S_Y(f) = S_X(f)\,|H(f)|^2 \qquad (8.81)$$

The PSD of the output is zero outside the band.

$$f_c - W/2 < f < f_c + W/2$$

The autocorrelation of the filter output is then

$$R_Y(t) = R_X(t) * h(t) * h(-t)$$

$$= \frac{N_0}{2}\, h(t) * h(-t)$$

the power is

$$P_Y = R_Y(0) = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \, df$$

$$= 2 \int_{f_c - W/2}^{f_c + W/2} \frac{N_0}{2} \, df$$

$$= N_0 W$$

$$= k T_c W.$$

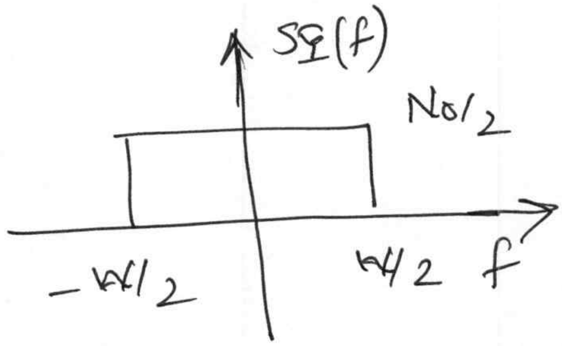
On the logarithmic scale:

$$P_Y = 10 \log_{10} (kTW) \text{ dBm}$$

The filtered noise ~~y~~ process $n(t)$ is bandpass and real valued and therefore has a baseband representation

$$n(t) = \underline{n_I(t)} \cos(2\pi f_c t) - \underline{n_Q(t)} \sin(2\pi f_c t)$$

lowpass processes of bandwidth $W/2$ centered at 0 with the PSD shown in Fig 3-13.



PSD of the quadrature components of Gaussian noise.

In the homework we established the identity:

$$\|x(t)\|^2 = \frac{1}{2}\left\{\|x_I(t)\|^2 + \|x_Q(t)\|^2\right\}$$
$$= \frac{1}{2}\|x_{bb}(t)\|^2$$

In the process of establishing this relationship (the power in the passband signal is half of ~~the~~ that in the baseband) we come across

$$\langle x_I, x_I\rangle = \langle x_Q, x_Q\rangle = \langle x(t), x(t)\rangle.$$

OR.

$$\|x_Q\|^2 = \|x_I\|^2 = \|x\|^2.$$

The above is for deterministic signals, for stochastic noise process $n(t)$ we need the operator $\mathbb{E}$.

$$\mathbb{E}\left[n_I^2(t)\right] = \mathbb{E}\left[n_Q^2(t)\right] = \mathbb{E}\left[n^2(t)\right]$$

$$= 2\int_0^{W/2} N_0 \, df$$

$$= N_0 W. \quad \text{— (A)}$$

The complex baseband ~~and~~ representation of the bandpass noise is:

$$n_{bb}(t) = n_I(t) + j n_Q(t)$$

$$= \left[n(t) + j\hat{n}(t)\right] e^{-j2\pi f_c t}.$$

(A) The power of bandlimited Gaussian noise depends linearly on bandwidth.

(B) Regardless of the actual environment temperature

290K (Average temperature of the Earth) is used as a reference.

At 290K, thermal noise will produce in one Hz of bandwidth a power equal to

$$kT_0 = 1.38065 \times 10^{-23} \times 290 = 4 \times 10^{-2} \text{ watts}$$

$$= -174 \text{ dBm}$$

In other words at $T_0 = 290K$ the one sided PSD of thermal noise is -174 dBm/Hz.

Noise-equivalent bandwidth: It is the bandwidth that will result in a flat PSD with the same total power:

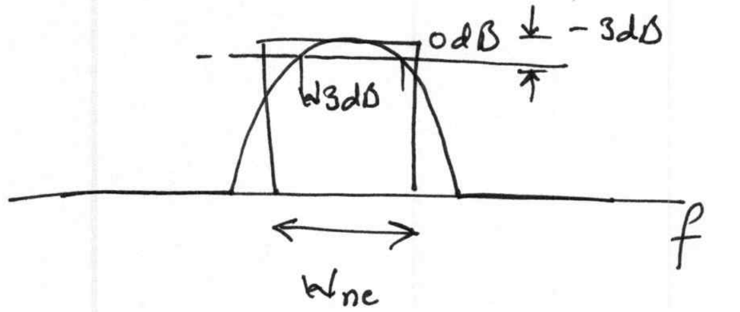


Illustration of noise equivalent bandwidth and 3-dB bandwidth.

We have that:

$$W_{ne} = \frac{1}{2\max|H(f)|^2}\int_{-\infty}^{+\infty}|H(f)|^2 df.$$

The power at the o/p is

$$P_Y = \frac{N_0}{2}\, 2W_{ne} \max|H(f)|^2$$

The definition is similar to the rectangular equivalent bandwidth.

"The noise equivalent bandwidth is the bandwith of the brickwall filter for which total power at the output is the same as the output power ~~at~~ of the actual real filter".

HW4 1

$\longrightarrow$ Example (page 86)

HW4 2

$\longrightarrow$ Example 8-2.

Some useful relationships: $\longrightarrow$.

$$SNR_{disp} = \frac{P_s + P_n}{P_n} = SNR + 1$$

$$\frac{\text{total power}}{\text{noise power}}$$

The relationship between actual power and displayed power is

$$SNR = 10 \log_{10} \left(10^{SNR_{disp}/10} - 1 \right)$$

- So far example we may have:

$$SNR_{disp} = 2dB,$$

It maybe that the actual signal is stronger than the noise.

Opposite to each each

However, from the above formula we realize that the actual signal SNR is $-2.3dB$.

In reality, the signal is not 2dB above but more than 2dB below the noise floor.

Only if the SNR is 15dB or greater then $SNR_{disp} \simeq SNR_{actual}$.

We will now take our knowledge of section 3-6 to 3-8 to cover (1) distortionless channels

(2) AWGN noise channels

(3) Passband Channels

(4) Baseband Channels

(5) Filter Structures Associated with them.

---

(1) Distortionless Channels: (Section 3-6)

- Distortionless channels highly desirable
  - → only constant delay
  - → flat attenuation.

Channel is distortionless

iff $y(t) = k\, x(t - t_d)$

k ↑ constant, t_d ↑ delay.

- Delay cannot be eliminated

  b/c EM waves travel at a finite speed.

  (3×10^8 m/s in vacuum)

  $\Downarrow$

  the delay is $\frac{1}{v_c} = d = 3.33\ \mu s/km$.

  in air.

  in other media the speed is lower

  e.g. 2.3×10^8 in copper wire

  2.0×10^8 in optical fiber

  slow in fibre because of its refractive index. $\Rightarrow$ delay is $5\ \mu s/km$.

Spectrum of an Ideal Channel:

The Fourier transform of the received signal is

$$\underline{Y}(f) = K\,\underline{X}(f)\,e^{-j2\pi f t_d}.$$

Since $\underline{Y}(f) = \underline{X}(f)\,H(f)$

it is clear that

$$K\,\underline{X}(f)\,e^{-j2\pi f t_d} = \underline{X}(f)\,H(f)$$

$$\Rightarrow \quad H(f) = K e^{-j2\pi f t_d}.$$

(1) The magnitude responsible of a distortionless channel is constant for all frequencies and the phase response is linear function of frequency:

$$|H(f)| = K$$

and $\quad \arg(H(f)) = -2\pi f t_d$

If the channel is distortionless, all frequencies components are attenuated by the same coefficient, i.e., the channel is not frequency-selective.

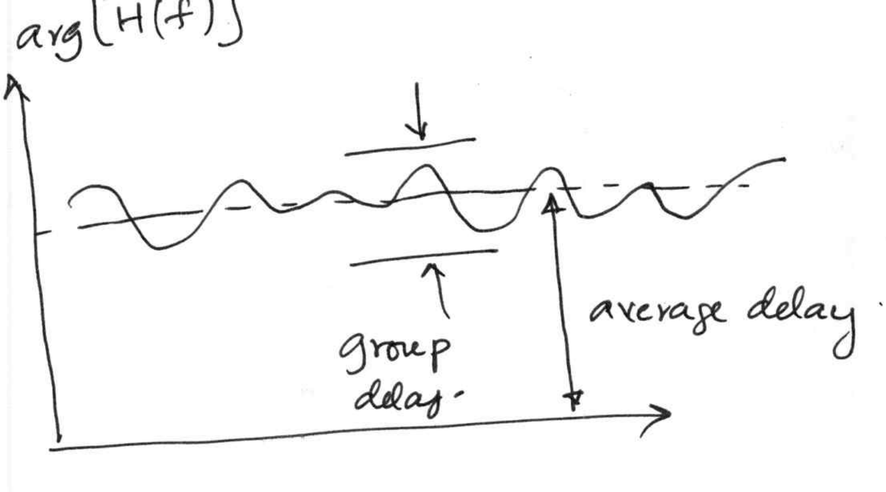
All frequency compoments of a signal are delayed when they pass through a channel.

- Phase delay is the time delay experienced by a sinusoidal signal as it ~~phas~~ passes through a channel.

$$\tau_d(f) = \frac{\arg(H(f))}{2\pi f}$$

- Group Delay: A more useful measure for modulated signals.

"Group delay is the amount of time it takes for a group of frequencies of a modulated signal to pass through the channel."



Real-world channels do not have distortionless response, only to an approximation — within the frequency span of the tx signal where there is constant gain and linear phase.

quiz.

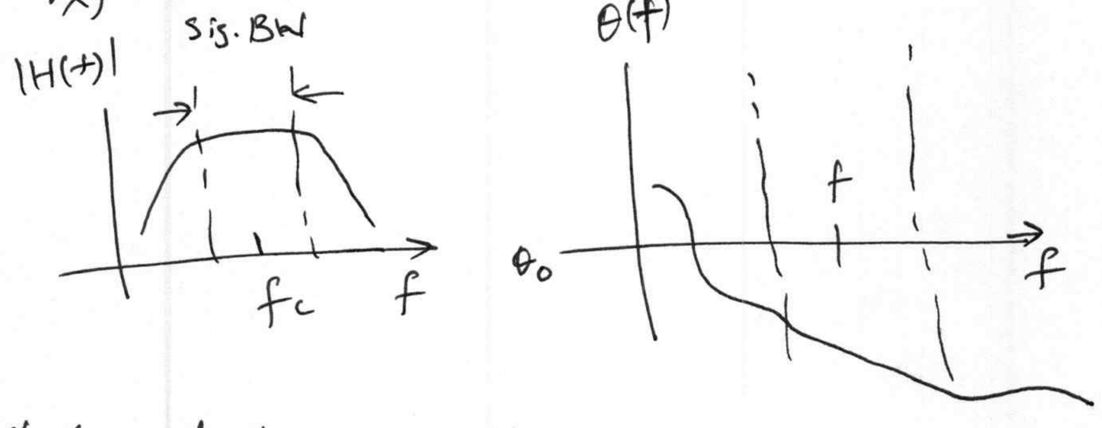
- Even if the channel is distortionless, there still can be <u>path loss</u>. — loss of signal strength as it travels through the medium.

- Path loss is defined as the ratio in dBm between tx and rx powers. (i/p and o/p w.r.t. to the channel).

$$A = 10 \log_{10} \frac{P_x}{P_y} = -20 \log_{10} K \text{ dB}.$$

- Path loss is alternatively called attenuation.

- In reality $K \ll 1$. (A fraction of tx is received at the rx).



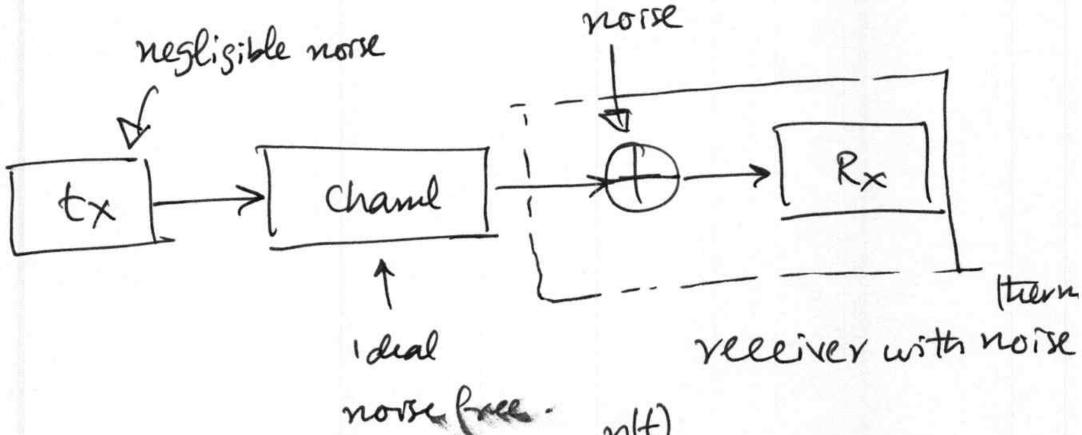
Magnitude and phase response

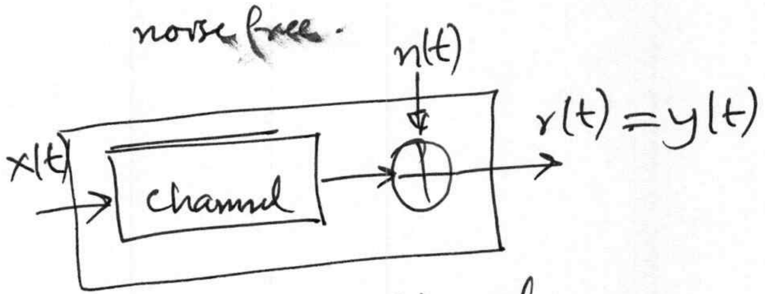
of ~~a in f~~ a practical channel with $H(f) = |H(f)| e^{j\theta(f)}$

The channel can be considered distortionless.

3-7 AWGN Channel:-

Additive White Gaussian Channel.





AWGN Chanel.

$$y(t) = r(t) = x(t) + n(t) \quad -(3.117)$$

where $n(t)$ is WSS Gaussian random process with

double-sided PSD $N_0/2$, at all frequencies.

- Practical comm systems are bandlimited:

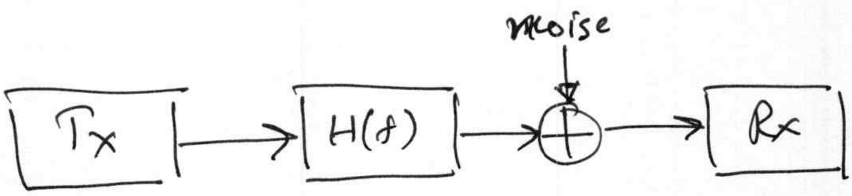
  The transmitted waveform $x(t)$ has a bandwidth of W Hz.

  - and a doublesided PSD of

$$\frac{P}{2W}$$

- Receivers are designed to filter out all frequencies, except the band of interest.

- The AWGN model is modified as follows:



So how do we represent this "band limited" version of AWGN?

⟶ In Chapter 2 that every continuous-time band limited signal can be sampled and

$$\left\{ \phi_k(t) : \phi_k(t) = \frac{1}{\sqrt{T_s}} \operatorname{sinc}\left[\frac{(t-kT_s)}{T_s} \right]$$

is an ONB for the signal space of bandlimited signals, where $T_s = \frac{1}{2W}$

Now, noise is bandlimited, it can be represented using this orthonormal basis:

$$y_k = \langle y(t), \phi_k(t) \rangle$$

$$= \langle x(t) + n(t), \phi_k(t) \rangle$$

$$= \langle x(t), \phi_k(t) \rangle + \langle n(t), \phi_k(t) \rangle$$

$$= x_k + n_k \qquad \text{——— (3.119)}$$

here y_k are the co-efficients of the expansion:

$$y(t) = \sum_k y_k \phi_k(t)$$

The n_k are <u>IID</u> random variables with zero mean and variance $\sigma^2 = N_0W$. Since the autocorrelation function $R_n(\tau) = 0$ for $\tau \neq 0$, two different noise samples are considered independent random variables.

Thus we have:

$$(3.117) \longrightarrow (3.119)$$

continuous time channel model | discrete time channel model.

to be continued .....